Observable Conditions for Robotaxi City Entry

This document organizes robotaxi city-entry conditions into three evidence categories:

1. **Publicly Confirmed Prerequisites or Enabling Factors**
2. **Operational Risks Demonstrated by Actual Events**
3. **Variables That Can Be Constructed from Public Datasets**

For part 2 and 3, the sources are news and datasets, so the variables are not official regulations for a robotaxi to be operated within a city; just derived variables that may be considered.

For each source, the structure is:

1. Paper, official document, or dataset and link
2. Summary of the source and its main findings
3. Variables that can be identified or constructed from the source

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Category 1. Publicly Confirmed Prerequisites or Enabling Factors

1. *A Framework for Automated Driving System Testable Cases and Scenarios*

Institution: U.S. Department of Transportation / National Highway Traffic Safety Administration
Document link: NHTSA PDF
Evidence type: Official federal technical framework

1. Summary and main findings

The document defines the **Operational Design Domain (ODD)** as the set of conditions under which an automated driving system is designed to operate.

The ODD includes:

- geographic area;
- roadway type;
- speed range;
- environmental and weather conditions;
- lighting conditions;
- traffic conditions;
- other operational constraints.

The document does not explain why a specific robotaxi company selected a specific city. It establishes the operating conditions that must be compatible with the ADS before deployment is technically feasible.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Possible measurement
Geographic operating-area compatibility	Whether the proposed service area is within the ADS-supported geographic domain	Confirmed technical prerequisite	Share of proposed service area covered by the authorized or validated geofence
Supported roadway-type share	Share of proposed roads belonging to roadway classes supported by the ADS	Confirmed technical prerequisite	Supported road miles divided by total proposed road miles
Supported speed-range share	Share of proposed roads within the ADS-supported speed range	Confirmed technical prerequisite	Compatible road miles divided by total proposed road miles

Variable	Meaning	Expected role in city entry	Possible measurement
Weather-condition compatibility	Whether local operating conditions fall inside the supported weather domain	Confirmed technical prerequisite	Share of planned service hours within supported weather conditions
Lighting and time-of-day compatibility	Whether operation is supported during daytime, nighttime, or particular lighting conditions	Confirmed technical prerequisite	Share of desired operating hours within supported lighting conditions
Traffic-condition compatibility	Whether local traffic conditions fall within the supported operating domain	Confirmed technical prerequisite	Share of operating periods within supported traffic conditions
Temporary-road-condition compatibility	Whether the ADS can handle temporary lane changes, work zones, and altered road geometry	Confirmed technical prerequisite	Share of road network or operating periods unaffected by unsupported temporary configurations

Source conclusion

> ODD compatibility is a necessary technical prerequisite for robotaxi deployment. Geography, roadway type, speed, weather, lighting, traffic, and temporary road conditions should be treated as separate feasibility conditions rather than combined into one subjective score.

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2. California DMV Autonomous Vehicle Programs

Institution: California Department of Motor Vehicles
Document link: California DMV — Autonomous Vehicles
Evidence type: Official state regulatory framework

1. Summary and main findings

The California DMV separates autonomous-vehicle activity into different authorization stages, including:

- testing with a safety driver;
- driverless testing;
- deployment.

This distinction shows that permission to test an AV does not automatically permit driverless deployment.

The DMV framework therefore provides direct evidence that regulatory authorization is a prerequisite for city entry.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Recommended variable form
Driverless testing legally permitted	Whether the operator can test without a human safety driver	Confirmed legal prerequisite	Binary indicator: 1/0
Driverless deployment legally permitted	Whether the operator can deploy driverless vehicles beyond testing	Confirmed legal prerequisite	Binary indicator: 1/0
Operator holds required permit	Whether the specific company has the permit required for the activity	Confirmed company-specific prerequisite	Binary indicator by company and jurisdiction
Authorized operating geography	Geographic limits stated in the permit	Confirmed legal-operational prerequisite	Service-area overlap ratio
Authorized operating conditions	Restrictions related to speed, roads, time, or other operating conditions	Confirmed legal-operational prerequisite	Number and type of permit restrictions

Source conclusion

> Driverless testing and driverless deployment must be treated as separate regulatory variables. A city cannot be considered fully feasible simply because autonomous testing is allowed.

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3. CPUC Autonomous Vehicle Passenger Service Programs

Institution: California Public Utilities Commission
Document link: CPUC — Autonomous Vehicle Programs
Evidence type: Official passenger-service regulatory framework

1. Summary and main findings

The CPUC regulates autonomous passenger service separately from vehicle testing and deployment.

The CPUC framework distinguishes pilot programs and deployment programs and imposes requirements related to:

- passenger-service authorization;
- safety plans;
- insurance;
- operational reporting;
- quarterly reporting.

This means that a robotaxi operator may be technically able to deploy vehicles but still lack authorization to provide paid passenger service.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Recommended variable form
Paid passenger service legally permitted	Whether the operator may carry paying passengers	Confirmed commercial prerequisite	Binary indicator: 1/0
Passenger-service authorization held	Whether the specific operator has CPUC authorization	Confirmed company-specific prerequisite	Binary indicator
Safety-plan requirement satisfied	Whether the required safety documentation has been submitted and accepted	Confirmed compliance prerequisite	Binary or categorical indicator
Insurance requirement satisfied	Whether the operator meets required insurance conditions	Confirmed compliance prerequisite	Binary indicator
Reporting requirement	Frequency and scope of required reporting	Confirmed compliance burden	Number of required reports or categories
Trip-data reporting availability	Whether deployment generates public operational data	Confirmed data-enabling factor	Binary indicator

Source conclusion

> Paid passenger-service authorization should be modeled separately from testing and deployment authorization. It is a direct commercial-entry prerequisite.

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6. DOE Alternative Fuels Data Center

Institution: U.S. Department of Energy
Dataset links:
- Alternative Fueling Station Locator
- AFDC Data Downloads
Evidence type: Official infrastructure dataset

1. Summary and main findings

The AFDC provides nationwide data on public charging infrastructure, including:

- station locations;
- charger types;
- connector counts;

- charging levels;
- available station attributes.

The dataset can be used to measure general EV charging readiness.

However, it does not show whether a robotaxi company has access to private charging depots, cleaning facilities, maintenance facilities, or fleet parking.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Evidence status
Public charging-station count	Number of public charging stations in the city or proposed geofence	General infrastructure proxy	Dataset-derived proxy
DC fast-charger count	Number of fast-charging facilities	Charging-readiness proxy	Dataset-derived proxy
Connector count	Potential simultaneous charging capacity	Public-capacity proxy	Dataset-derived proxy
Charger power	Potential charging speed	Charging-duration proxy	Dataset-derived proxy
Distance from service area to chargers	Deadheading required to reach public charging	Operational-cost proxy	Dataset-derived proxy

Source conclusion

> AFDC data can measure general EV charging infrastructure, but it cannot be treated as direct evidence of proprietary robotaxi depot capacity.

Category 2. Operational Risks Demonstrated by Actual Events

1. Reuters — Waymo Recall over Flooded-Road Risk

Source: Reuters
Article link: Waymo recall over flooded-road risk
Evidence type: Reported real-world robotaxi operating event based on Waymo and NHTSA information

1. Summary and main findings

Waymo recalled approximately 3,800 robotaxis after a driverless vehicle entered a flooded lane during extreme weather.

The reported response included:

- stronger weather-related restrictions;
- reduced operating scope;
- restrictions on access to flash-flood-prone areas;
- software and map-related corrective actions.

This event provides direct evidence that flooding and severe weather can reduce robotaxi service availability.

It does not establish that Waymo uses a specific rainfall or FEMA threshold when selecting cities.

2. Variables identified from the source

Variable	Observed operational effect	Expected role in city-entry analysis	Evidence status
Flooded-road exposure	Can make routes unsafe or unusable	Operational-risk factor	Demonstrated by actual event
Heavy-rain exposure	Can trigger operating restrictions	Operational availability factor	Demonstrated by actual event
Flash-flood exposure	Can cause route or area exclusion	Service-area reliability factor	Demonstrated by actual event

Variable	Observed operational effect	Expected role in city-entry analysis	Evidence status
Weather-related service restriction	Reduces operating hours and effective fleet	Scenario-dependent capacity factor	Demonstrated by actual event
Hazard-related map or route update	Can temporarily change the geofence or route set	Dynamic feasibility factor	Demonstrated by actual event

Source conclusion

> Flooding, heavy rain, and flash-flood exposure are demonstrated operational risks. They should be modeled as factors that reduce service availability, not as publicly confirmed corporate city-selection thresholds.

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2. USDOT Work Zone Data Exchange

Institution: U.S. Department of Transportation
Dataset link: Work Zone Data Exchange
Evidence type: Official temporary-road-condition dataset

1. Summary and main findings

The Work Zone Data Exchange standard provides data related to active work zones, such as:

- work-zone location;
- lane closure;
- temporary road changes;
- start and end times;
- road restrictions.

Construction zones change lane geometry, road markings, traffic control, and expected driving behavior.

The dataset does not prove that a company uses work-zone frequency as a formal city-selection criterion. It provides a way to measure exposure to temporary road complexity.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Evidence status
Active work-zone count	Number of active work zones in the proposed service area	Operational-risk proxy	Dataset-derived
Work-zone density	Work zones per road mile or square mile	Temporary-road-complexity proxy	Dataset-derived
Lane-closure hours	Total duration of lane closures	Route-capacity risk	Dataset-derived
Work-zone duration	Length of time a road is affected	Service-reliability risk	Dataset-derived
Share of major routes affected	Portion of key corridors affected by construction	Geofence-connectivity risk	Dataset-derived

Source conclusion

> Work-zone data can quantify exposure to temporary road configurations. It is best treated as an operational-risk proxy rather than a confirmed city-selection rule.

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3. FHWA Highway Performance Monitoring System

Institution: Federal Highway Administration
Dataset link: FHWA HPMS
Evidence type: Official road-network dataset

1. Summary and main findings

HPMS provides nationwide information on:

- functional road classification;
- road mileage;
- traffic volume;
- roadway characteristics;
- highway-system structure.

These data can be used to evaluate whether a proposed service area depends heavily on road types that the ADS may not support.

The dataset itself does not prove that robotaxi companies select cities based on a particular freeway-share threshold.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Evidence status
Freeway-road share	Share of road miles classified as freeway or interstate	Conditional operational factor	Dataset-derived proxy
Arterial-road share	Share of major arterial roads	ODD compatibility proxy	Dataset-derived proxy
Local-road share	Share of local roads	ODD compatibility proxy	Dataset-derived proxy
Traffic volume	AADT or related traffic measures	Operational-load proxy	Dataset-derived proxy
Road mileage by functional class	Network composition	Road-type compatibility measure	Dataset-derived proxy

Source conclusion

> Road-network composition can be measured through HPMS. Freeway dependence should be interpreted only in combination with the specific ADS's freeway capability.

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4. NOAA Climate Data Online and Storm Events Database

Institution: National Oceanic and Atmospheric Administration
Dataset links:
- NOAA Climate Data Online
- NOAA Storm Events Database
Evidence type: Official weather and hazard datasets

1. Summary and main findings

NOAA provides historical weather and hazard data, including:

- precipitation;
- snowfall;
- temperature;
- visibility;
- severe storms;
- flash floods;
- other recorded weather events.

These data can be used to compare the exposure of candidate cities to weather conditions that may fall outside a robotaxi system's ODD.

The datasets do not independently establish the weather limits of any specific ADS.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Evidence status
Heavy-rain days or hours	Frequency of intense precipitation	Weather-risk proxy	Dataset-derived
Snow and ice days	Frequency of snow, ice, or freezing conditions	ODD exposure proxy	Dataset-derived
Low-visibility hours	Frequency of visibility below a threshold	ODD exposure proxy	Dataset-derived
Flash-flood frequency	Historical flash-flood events	Demonstrated flood-risk proxy	Dataset-derived
Severe-weather days	Frequency of major weather events	Service-interruption proxy	Dataset-derived

Source conclusion

> NOAA data can quantify weather exposure, but company-specific ODD thresholds are still required to determine whether a condition is actually unsupported.

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5. FEMA National Flood Hazard Layer and National Risk Index

Institution: Federal Emergency Management Agency
Dataset links:
- FEMA National Flood Hazard Layer
- FEMA National Risk Index
Evidence type: Official hazard datasets

1. Summary and main findings

FEMA provides spatial data on flood hazard areas and broader hazard-risk indicators.

These datasets can be combined with road-network data to estimate how much of a proposed robotaxi service area is exposed to flooding.

The datasets do not establish that a company formally uses FEMA risk scores in its city-selection process.

2. Variables identified from the source

Variable	Meaning	Expected role in city entry	Evidence status
Flood-exposed road share	Share of road miles overlapping FEMA flood zones	Demonstrated operational-risk proxy	Dataset-derived
Flood-exposed service-area share	Share of proposed geofence in flood hazard areas	Service-area reliability proxy	Dataset-derived
Hazard expected annual loss	FEMA estimate of long-run hazard exposure	General risk proxy	Dataset-derived
Number of critical corridors in flood zones	Count of key routes exposed to flooding	Network-resilience proxy	Dataset-derived

Source conclusion

> FEMA data can measure geographic exposure to flooding and other hazards. These measures are researcher-constructed proxies for documented operational risks.

Category 3. Variables That Can Be Inferred or Constructed from Datasets

This category combines variables that can be operationalized across candidate cities using official datasets.

These variables are not publicly confirmed as internal corporate city-selection criteria.

1. Regulatory datasets

1. Potential sources:

- state DMV autonomous-vehicle permit records;
- state public-utility commission records;
- state transportation-agency AV laws and rules.

2. Summary and use

Regulatory datasets can be used to compare whether cities or states permit:

- driverless testing;
- driverless deployment;
- paid passenger service;
- commercial operation;
- specific operating geographies.

3. Variables that can be constructed

Variable	Construction method	Recommended interpretation
Driverless testing eligibility	Binary indicator from state rules	Confirmed prerequisite
Driverless deployment eligibility	Binary indicator from state rules	Confirmed prerequisite
Paid passenger-service eligibility	Binary indicator from state rules	Confirmed prerequisite
Number of active AV permit holders	Count by jurisdiction	Market-readiness proxy
Regulatory reporting burden	Count or category of reporting requirements	Compliance-cost proxy
Regulatory approval duration	Time between application and authorization, where available	Entry-delay proxy

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2. Road-network datasets

1. Potential sources:

- FHWA HPMS;
- local DOT road GIS;
- EPA Smart Location Database;
- WZDx.

2. Summary and use

Road datasets can be used to measure whether a proposed service area is compatible with the ADS and how complex the operating environment may be.

3. Variables that can be constructed

Variable	Construction method	Recommended interpretation
Supported roadway-type share	Supported road miles divided by total proposed road miles	Direct ODD compatibility measure
Supported speed-range share	Compatible road miles divided by total proposed road miles	Direct ODD compatibility measure
Freeway dependence	Freeway miles or freeway-dependent OD trips divided by total	Conditional proxy
Intersection density	Intersections per square mile	Road-complexity proxy

Variable	Construction method	Recommended interpretation
Work-zone exposure	Work zones or closure hours per road mile	Temporary-complexity proxy
Traffic volume	Average traffic flow on proposed service roads	Operational-load proxy
Travel-time reliability	Variation in travel time or Planning Time Index	Fleet-productivity proxy

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3. Weather and hazard datasets

1. Potential sources:

- NOAA Climate Data Online;
- NOAA Storm Events Database;
- FEMA National Flood Hazard Layer;
- FEMA National Risk Index.

2. Summary and use

Weather and hazard data can be used to measure exposure to operating conditions that have been documented as robotaxi risks or that may fall outside an ADS's ODD.

3. Variables that can be constructed

Variable	Construction method	Recommended interpretation
Heavy-rain exposure	Annual days or hours above a defined rainfall threshold	Weather-risk proxy
Snow and ice exposure	Annual snow, ice, or freezing-condition days	ODD exposure proxy
Low-visibility exposure	Annual hours below a visibility threshold	ODD exposure proxy
Flash-flood frequency	Historical event count per year	Demonstrated-risk proxy
Flood-exposed road share	Road miles inside FEMA flood zones divided by total road miles	Demonstrated-risk proxy
Severe-weather exposure	Annual severe-weather events or hours	Service-interruption proxy

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4. Charging and depot-readiness datasets

1. Potential sources:

- DOE AFDC;
- local parcel and zoning data;
- commercial property or warehouse-cost data;
- local utility interconnection data.

2. Summary and use

These datasets can be used to approximate whether a candidate city has sufficient infrastructure to support robotaxi charging, parking, maintenance, and cleaning.

3. Variables that can be constructed

Variable	Construction method	Recommended interpretation
Public fast-charger density	DC fast chargers per square mile or per 100,000 residents	General charging-readiness proxy
Connector capacity	Number of public charging connectors	Public-capacity proxy
Distance to charging infrastructure	Mean distance from proposed geofence to chargers	Charging deadheading proxy

Variable	Construction method	Recommended interpretation
Commercial land cost	Lease or property cost for potential depot areas	Depot-cost proxy
Suitable depot-parcel availability	Count or area of parcels meeting minimum requirements	Depot-feasibility proxy
Grid or utility readiness	Available interconnection or charging capacity, where public	Infrastructure-readiness proxy

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5. Prior AV activity and market-readiness data

1. Potential sources:

- company announcements;
- state permit records;
- local pilot-program records;
- company service maps.

2. Summary and use

Prior testing, pilot activity, and service announcements can be used to measure how far a city has progressed toward commercial deployment.

3. Variables that can be constructed

Variable	Construction method	Recommended interpretation
Prior AV testing presence	Binary indicator	Enabling-factor proxy
Testing duration	Months or years since first local test	Validation-maturity proxy
Existing public-rider pilot	Binary indicator	Confirmed readiness factor
Announced future market	Binary indicator	Confirmed expansion status
Existing local partnership	Binary indicator	Confirmed enabling factor
Existing service-area size	Square miles or population covered	Operating-footprint measure

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Final Variables for Research

A. Publicly confirmed prerequisites or enabling factors

1. Driverless testing legally allowed
2. Driverless deployment legally allowed
3. Paid passenger service legally allowed
4. Authorized geographic service area
5. Supported roadway-type share
6. Supported speed-range share
7. Weather and lighting compatibility with the ODD
8. Prior local testing or pilot activity
9. Local operating or platform partnership

10. Feasible initial geofence
11. Feasible depot and charging plan

B. Operational risks demonstrated by actual events

1. Heavy-rain exposure
2. Flooded-road exposure
3. Flash-flood exposure
4. Construction-zone exposure
5. Temporary road-closure exposure
6. Unsupported freeway dependence
7. Weather-related service-interruption risk

C. Dataset-derived proxy variables

1. Flood-exposed road share
2. Heavy-rain days or hours
3. Snow and ice days
4. Low-visibility hours
5. Work-zone density
6. Lane-closure hours
7. Freeway-road share
8. Intersection density
9. Traffic volume
10. Travel-time reliability
11. Public fast-charger density
12. Distance to charging infrastructure
13. Commercial land or warehouse cost
14. Number of active AV permit holders
15. Historical AV testing duration